

NAME: _____ CLASS: _____ INDEX: _____



CATHOLIC JUNIOR COLLEGE
JC2 PRELIM EXAMINATION
Higher 1

BIOLOGY

Paper 1 Multiple Choice

8876/01

28 AUGUST 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write and/or shade your name, NRIC / FIN number and HT group on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft 2B pencil** on the separate Answer Sheet.

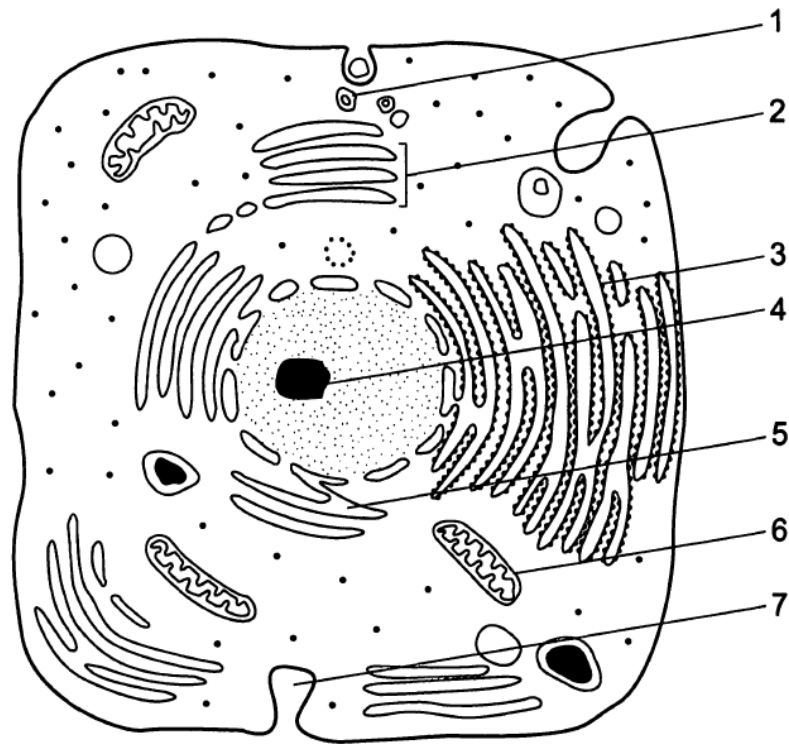
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

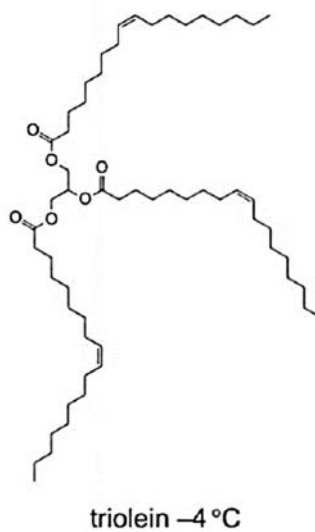
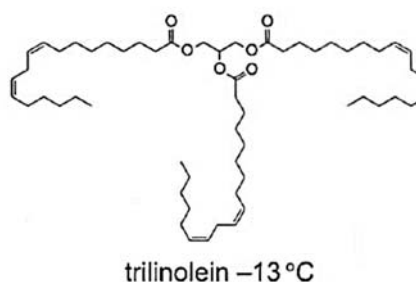
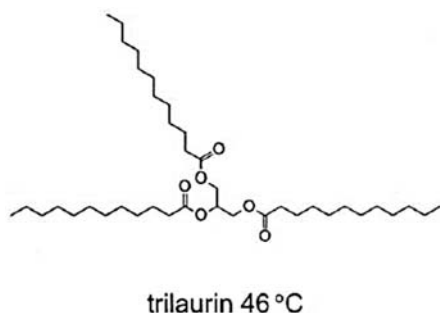
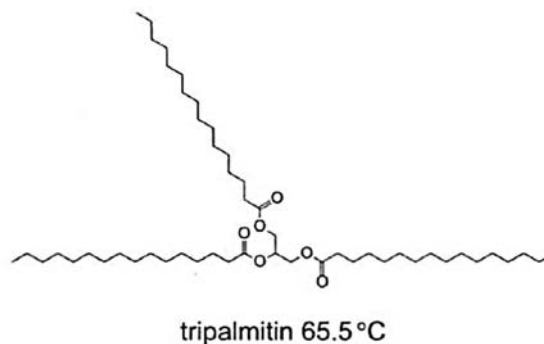
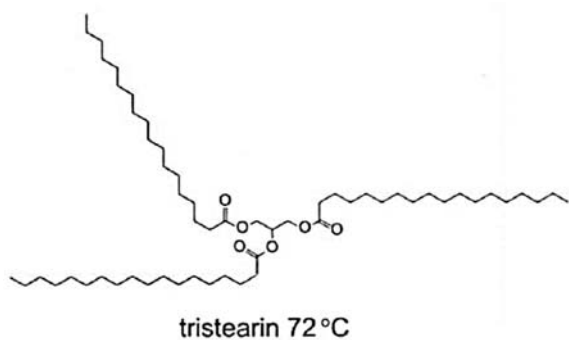
- 1 The diagram shows a section of a generalised animal cell as seen under the electron microscope.



Where are the proteins and lipids synthesised and transported, packaged and secreted?

	synthesised and transport		packaged	secreted
	proteins	lipid	proteins and lipids	proteins and lipids
A	3	5	2	1
B	4	6	3	7
C	5	3	4	7
D	6	1	2	5

- 2 The formulae and melting points of five triglycerides are shown in the diagram. Each triglyceride contains three identical fatty acids.

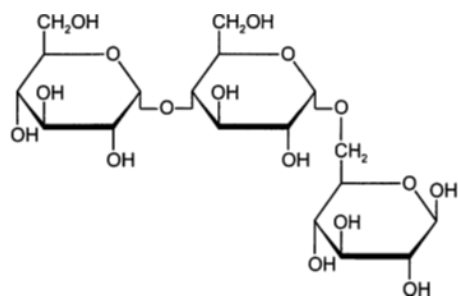


Which two structural features of the molecules make the melting point higher?

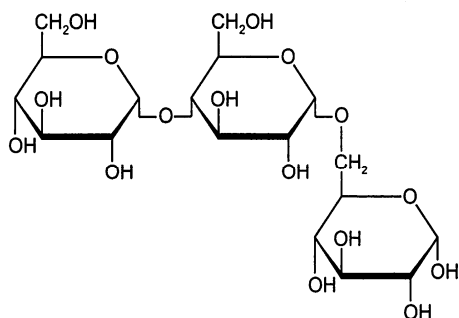
	number of double bonds	length of fatty acid chains
A	fewer	shorter
B	fewer	longer
C	more	longer
D	more	shorter

- 3 Which diagram correctly shows a trisaccharide containing both 1,4-glycosidic and 1,6-glycosidic bonds, and formed from both α -glucose and β -glucose molecules?

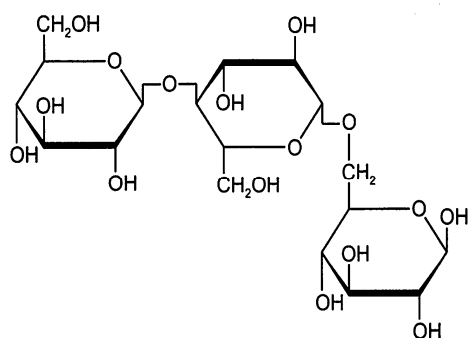
A



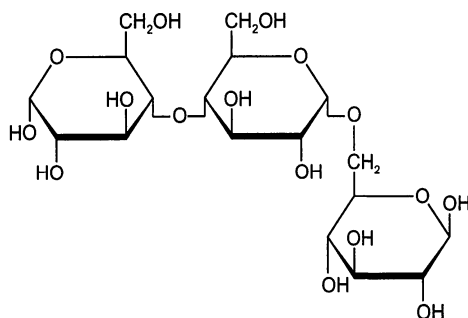
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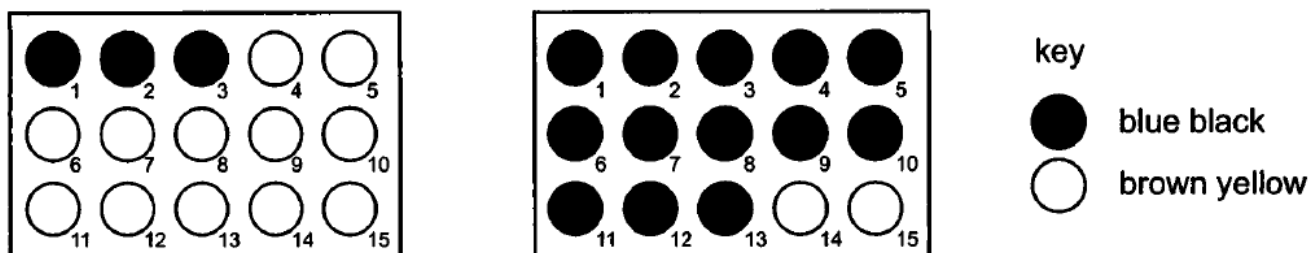
C



D



- 4 An experiment was carried out to investigate the digestion of starch using amylase at two different temperatures. A sample was removed from each mixture at 15 second intervals and placed onto a spotting tile well containing two drops of iodine in KI solution. The results are shown in the diagram.



Which shows the correct temperatures and times for the complete digestion of starch?

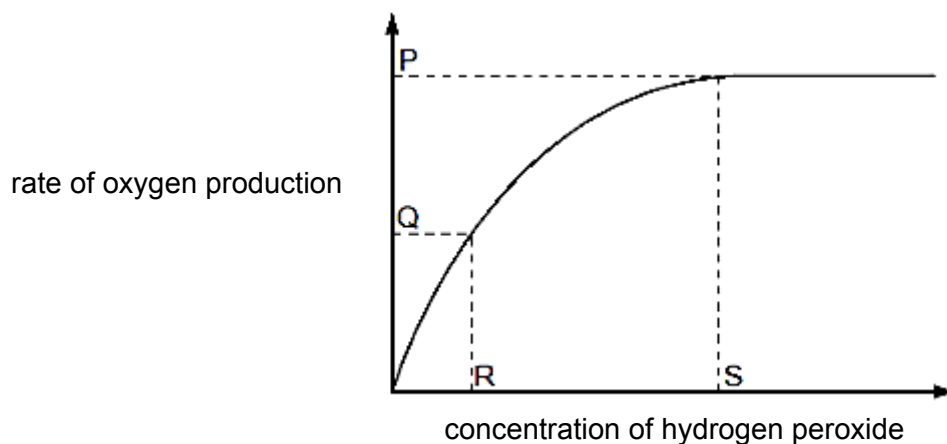
	temperature / °C	time / s
A	10	0.45
	30	3.15
B	10	195
	30	45
C	10	45
	30	195
D	10	3.15
	30	0.45

- 5 The diagram shows the action of a liver enzyme called catalase, which breaks down hydrogen peroxide into water and oxygen.



The rate of this reaction can be determined by measuring the volume of oxygen produced in a given length of time. Students added small cubes of fresh liver tissue to hydrogen peroxide solution of varying concentrations and measured the volume of oxygen produced.

The graph shows how the concentration of hydrogen peroxide affected the rate of oxygen production.



Which statements are correct?

- 1 At P, the rate of reaction is at its maximum.
- 2 At Q, all of the enzyme active sites are occupied by substrate molecules.
- 3 At Q, the rate of reaction is limited by the concentration of the substrate.
- 4 At S, all of the enzyme active sites are occupied by substrate molecules.

- A** 1 and 4
B 2 and 4
C 1, 2 and 3
D 1, 3 and 4

- 6 The concentration of glucose is higher in the blood plasma.

By which process does glucose move into red blood cells from the blood plasma?

- A** active transport
B endocytosis
C facilitated diffusion
D osmosis

- 7** It has been found that stem cells transferred from the intestinal lining to the bone marrow produce all of the different types of blood cell instead of intestinal cells.

Which statement explains this?

- A** All stem cells are totipotent.
- B** Environmental factors change the expression of specific genes.
- C** Specific genes are destroyed by endonucleases.
- D** Specific genes are hidden by condensation of some chromosomes.

- 8** Blood transfusion laboratories around the world are hoping to produce large numbers of red blood cells (rbcs) from 'spare' human embryos produced during *in vitro* fertilisation procedures.

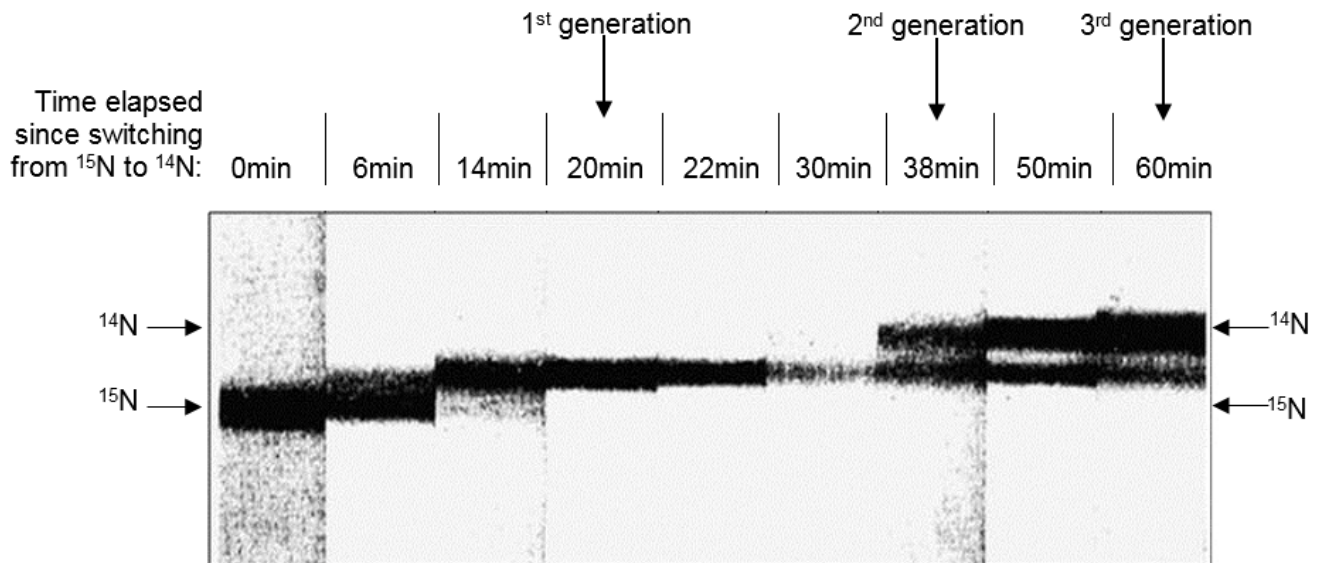
Embryonic stem cells are removed from an embryo and cultured in a growth medium that stimulates their differentiation into rbcs.

Which statement correctly describes this differentiation?

- A** Multipotent embryonic stem cells differentiate into pluripotent blood stem cells and then into rbcs.
- B** Pluripotent embryonic stem cells differentiate into multipotent blood stem cells and then into rbcs.
- C** Totipotent embryonic stem cells differentiate into multipotent blood stem cells and then into rbcs.
- D** Totipotent embryonic stem cells differentiate into pluripotent blood stem cells and then into rbcs.

- 9 In the classic paper that demonstrated the semi-conservative replication of DNA, scientists Meselson and Stahl began by showing that DNA itself will form a band when subjected to density gradient centrifugation.

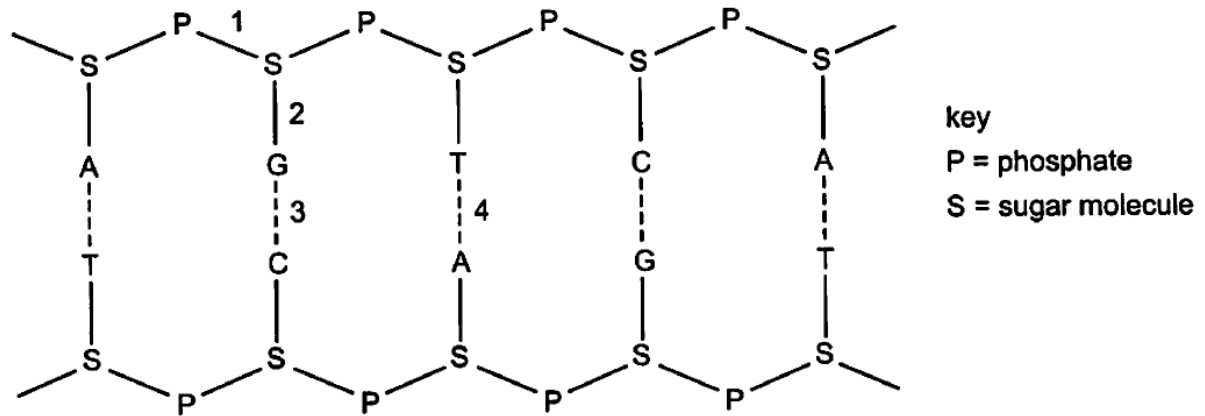
Escherichia coli grown in ^{15}N DNA were switched to ^{14}N and then harvested at eight different time points. The DNA was centrifuged resulting in the banding pattern shown.



Which statements correctly explain the results?

- 1 At 20 min, the entire DNA of *E. coli* exists as hybrid with 100% ^{15}N DNA.
 - 2 At 20 min, DNA of *E. coli* is 50% hybrid with 50% ^{15}N DNA.
 - 3 At 38 min, there are two bands consisting of 50% hybrid DNA and 50% light DNA.
 - 4 At 60 min, there is 25% hybrid DNA and 75% light DNA.
- A** 1 and 2
B 3 and 4
C 2, 3 and 4
D 1, 2, 3 and 4

- 10 The diagram shows part of a nucleic acid.



Which row correctly describes the bonds shown in the diagram at positions 1, 2, 3 and 4?

	is formed by condensation	forms a di-ester	occurs during transcription	involves attraction between polar molecules
A	1	2	1, 2 and 3	3
B	1 and 2	1	3 and 4	3 and 4
C	2, 3 and 4	1	3 and 4	1, 3 and 4
D	2	3	1, 3 and 4	4

- 11 The same length of DNA in a eukaryote can code for more than one protein.

When are different introns removed in order to allow the production of different mRNAs?

- A** at transcription
- B** before transcription
- C** before translation
- D** during translation

- 12** The active messenger RNAs (active mRNAs) in tissue cells can be isolated by passing the homogenised cell contents through a fractionating column. The column has short lengths of uracil nucleotides attached to a solid supporting material. Most molecules of mRNA that pass through the column quickly break up into small pieces and cannot be translated

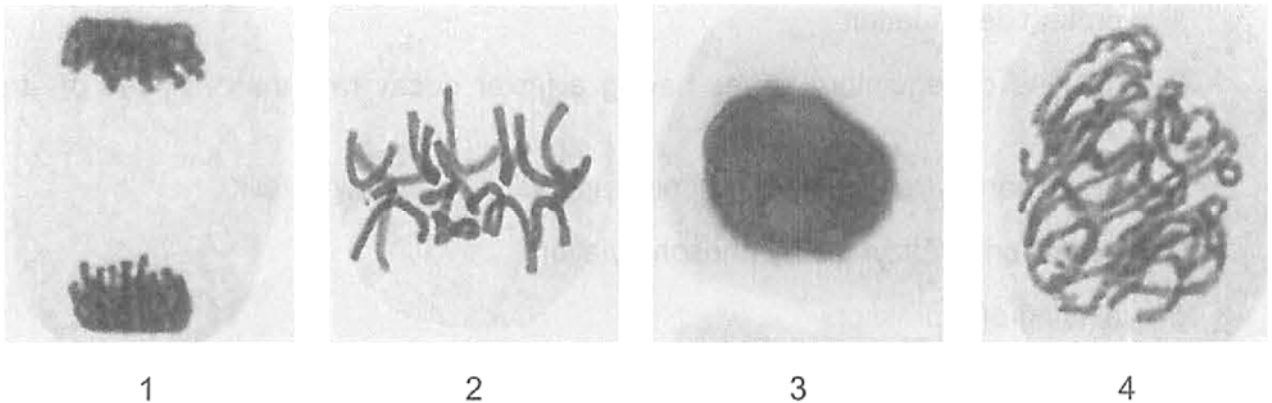
The active mRNAs that attach to the column can be separated again by appropriate treatment.

Which statements correctly describe active mRNA?

- 1 Active mRNAs are held to the fractionating column by bonds between adenine and uracil bases.
- 2 Active mRNAs can be released from fractionating column by breaking hydrogen bonds.
- 3 Only mRNAs with polyadenine tailing can be translated.
- 4 Polyadenine tailing stabilizes mRNA and prevents it from being broken up.

- A** 1 and 2
B 1, 2 and 3
C 3 and 4
D 1, 2, 3 and 4

- 13** The photomicrographs show different stages of the mitotic cell cycle.



Which row matches the name of a stage, a description of some of the events happening at this stage and a photomicrograph of a stage of the mitotic cell cycle?

	name of stage	description	photomicrograph
A	anaphase	Centromeres bind to spindle microtubules between centrioles. Chromosomes are pulled into position.	1
B	metaphase	Chromosomes align at equator of cell. Microtubules begin to pull the two parts of each chromosome in opposite directions.	2
C	prophase	Chromatin condenses. Nuclear envelope disperses. Nucleolus no longer visible.	3
D	telophase	Chromosomes become less condensed. Spindle microtubules disperse.	4

- 14 Fig. 14.1 represents the changes in the quantity of DNA in two types of cell divisions that occur in different types of cells of an organism. Fig. 14.2 shows the entire set of homologous chromosomes in a diploid cell of this organism before it undergoes the type of nuclear division that leads to **P**.

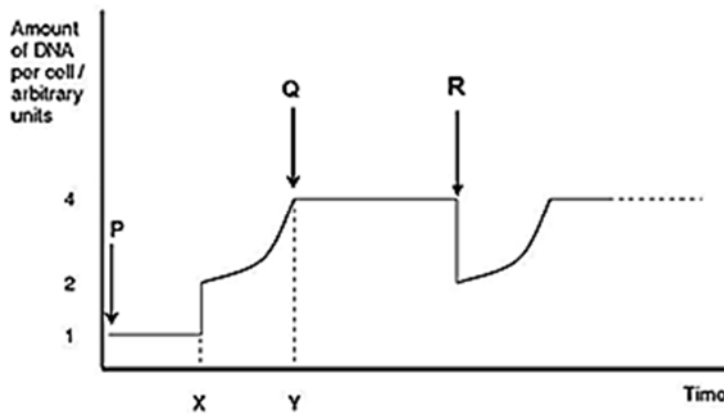


Fig. 14.1

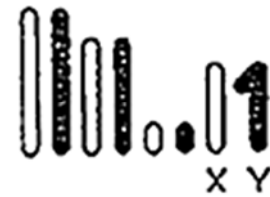


Fig. 14.2

Identify the correct combination of outcomes within a cell in this organism at **P**, **Q** and **R**.

	At P	At Q	At R
A		Diploid set of homologous chromosomes, each with identical sister chromatids.	Diploid set of homologous chromosomes, each a single DNA molecule.
B		Diploid set of homologous chromosomes, each with identical sister chromatids.	Haploid set of chromosomes, each a single DNA molecule.
C		Diploid set of homologous chromosomes, each a single DNA molecule.	Haploid set of chromosomes, each a single DNA molecule.
D		Tetraploid sets of homologous chromosomes, each a single DNA molecule.	Diploid set of homologous chromosomes, each a single DNA molecule.

- 15 A toxic chemical causes malfunction of the centrioles in animal cells.

Which process in meiosis is likely to be directly affected by the chemical?

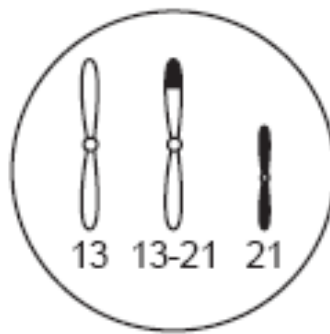
- A** crossing over between homologous chromosomes
- B** migration of chromosomes to opposite poles of the cell
- C** pairing of homologous chromosomes
- D** replication of centromeres

16 Which set of terms matches the definitions in the table?

	definition			
	the structure that replicates in the S phase	in animal cells, the 'pinching in' process that divides the cytoplasm	the cell structure that disassembles to allow chromosome attachment to the spindle	the phase of the cell cycle immediately prior to entering mitosis
A	centriole	cytokinesis	nuclear envelope	S phase
B	centriole	late telophase	nucleolus	S phase
C	chromatid	cytokinesis	nuclear envelope	G ₂ phase
D	chromatid	late telophase	nucleolus	G ₂ phase

17 Down's syndrome can be caused by a trisomy of chromosome 21, but can also result from the translocation of chromosome 21 into chromosome 13, forming a single chromosome 13-21.

The diagram shows chromosomes 13 and 21 in the nucleus of a diploid (2n) testis cell from a phenotypically normal male carrier of a 13-21 translocation. This cell has a chromosome number of 45.



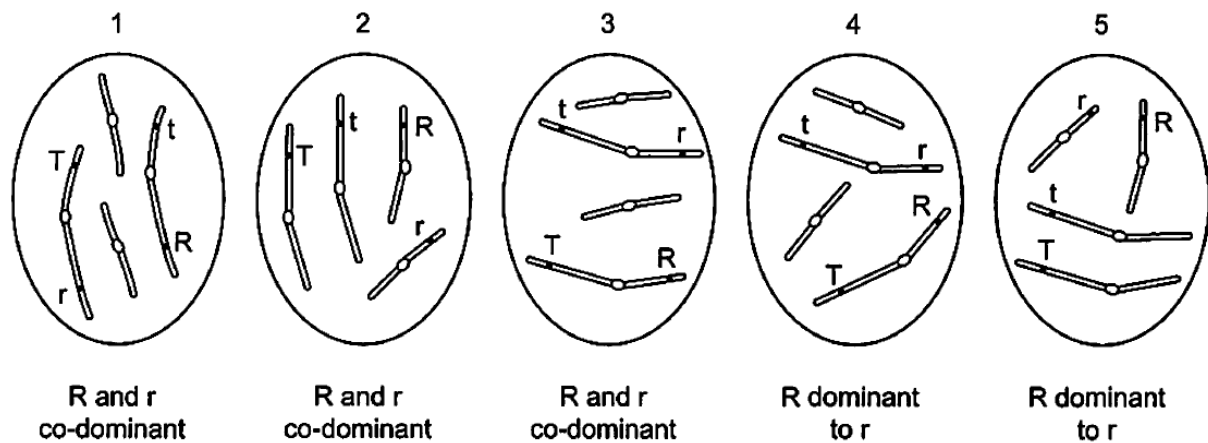
Which is **not** a likely outcome of fertilisation of normal oocytes by sperm from this male?

	chromosomes in sperm	embryo
A	13 and 21	2n =46 normal phenotype
B	13-21	2n =45 normal phenotype
C	13-21 and 21	2n =46 Down's syndrome
D	13-21 and 21	2n =47 Down's syndrome

- 18** In a family of flowering plants, height is controlled by a pair of alleles. The allele for tall (T) is always dominant to the allele for short (t). The flower colour is also controlled by a pair of alleles.

In some species, the allele for red (R) is dominant to the allele for white (r). In other species the colour alleles are co-dominant. (For simplicity, the symbols R and r are used for the co-dominant alleles.)

The diagram shows the chromosome arrangement and information about the height alleles and the flower colour allele in five species of this family of plants.



Each of the plants 1, 2, 3, 4 and 5 was test crossed.

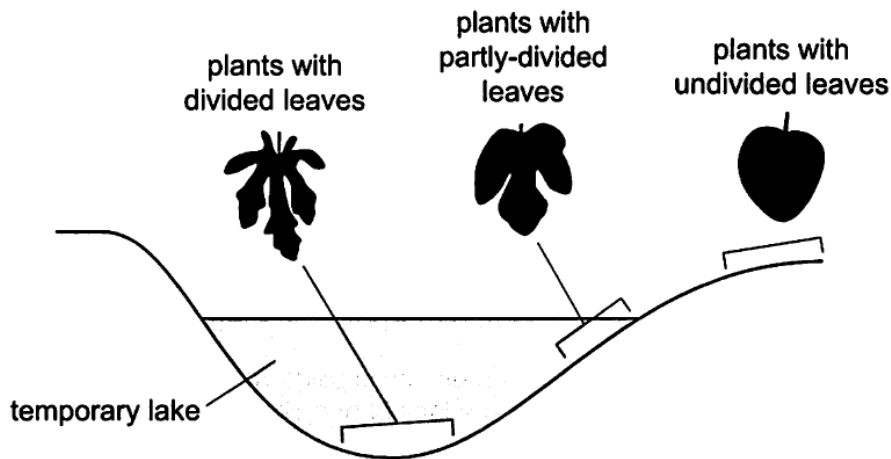
Assuming there is no crossing over, which plants would produce offspring with the phenotypes short with white flowers and tall with red flowers?

- A** 1,3 and 4
B 2,4 and 5
C 2 and 5 only
D 4 and 5 only
- 19** In a monohybrid, X-linked (sex-linked) genetic cross involving two alleles, dominance and recessiveness is observed in the phenotypes obtained. The link between genotype and phenotype is not always obvious when the parental and offspring phenotypes are recorded.

What is the best explanation of this observation?

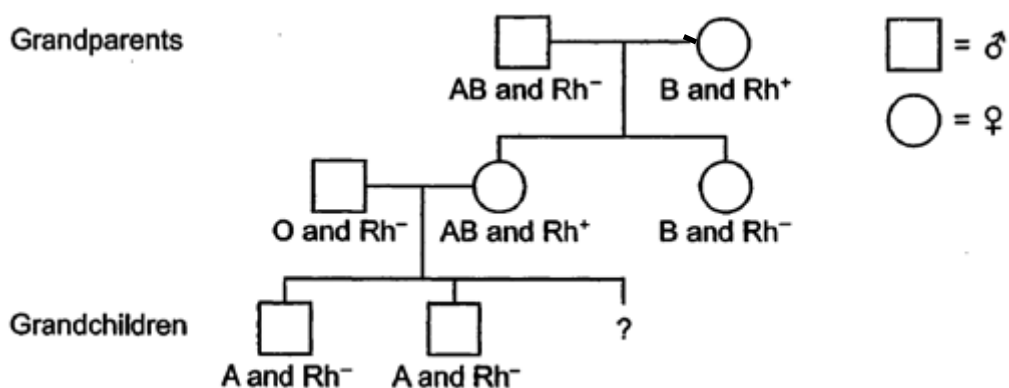
- A** The dominant and recessive phenotypes can be explained by the transcription of different nucleotide sequences to produce two different mRNA molecules, which produce a functional and a non-functional protein.
- B** The expression of the recessive allele only occurs when it is in the male, because the Y chromosome is lacking the entire sequence of nucleotides corresponding to the production of a different protein.
- C** The nucleotide sequences of the alleles occurring at different loci leads to the production of active and inactive enzymes, so that the heterozygous phenotype only has half of the enzyme concentration of the homozygous dominant phenotype.
- D** The recessive allele present in the male is unlikely to be expressed in the same way as it would if it occurred in the female heterozygote, as the male does not possess a corresponding nucleotide sequence on the X chromosome.

- 20 The diagram shows the distribution of plants with different leaf shapes in an area where flooding has caused the development of a temporary lake. The plants are all of the same species. When there is no flooding, all the leaves are undivided.



What could explain these differences in leaf shape during flooding?

- A changes in gene expression dependent on the soil water content
 - B cross-fertilisation between plants with different leaf shapes
 - C random assortment of alleles in the plant population
 - D self-fertilisation allowing adaptation to the local environment
- 21 The diagram shows a family tree.



What is the probability that the third grandchild will be a boy with blood group B and Rh positive blood?

- A 0.0625 (1 in 16)
- B 0.125 (1 in 8)
- C 0.25 (1 in 4)
- D 0.5 (1 in 2)

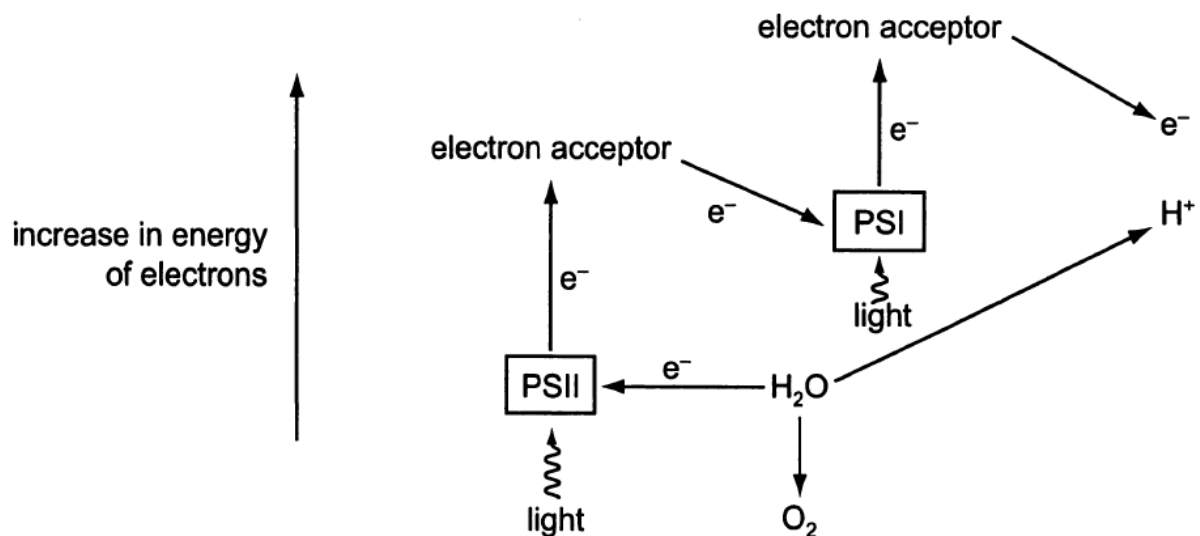
22 What describes oxidative phosphorylation?

- A addition of phosphate to ADP using energy gained by transferring electrons along a chain of carriers
- B addition of phosphate to ADP using energy gained by transferring electrons between chlorophyll molecules
- C addition of phosphate to glucose in the first step of glycolysis
- D removal of phosphate from ATP with the release of energy for work within the cell

23 Where does ethanol formation occur in a yeast cell?

- A cell vacuole
- B cytoplasm
- C Golgi apparatus
- D mitochondrion

24 The diagram shows the path taken by electrons and the formation of hydrogen ions in the light-dependent stages of photosynthesis.

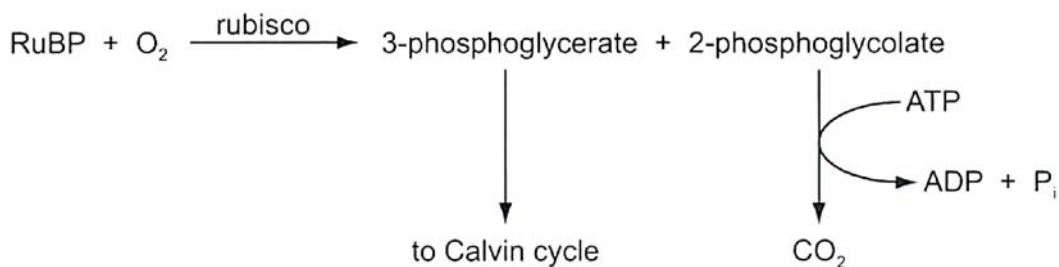


What are the electrons and hydrogen ions used to produce?

- A ATP from ADP
- B ATP from ADP and reduced NADP from NADP
- C glycerate 3-phosphate from glyceraldehyde 3-phosphate
- D reduced NADP from NADP

- 25** Rubisco is the carbon dioxide-fixing enzyme. One rubisco molecule has eight active sites where carbon dioxide fixation occurs, with each active site catalysing only three reduction reactions per second.

The enzyme also catalyses, at the same active sites, the addition of oxygen to ribulose biphosphate (RuBP). This reaction is favoured when oxygen concentrations in the leaf are high and carbon dioxide concentrations are low.



Which of the facts is paired with a correct explanation?

	fact	explanation
A	On very hot, dry days stomata close to prevent water loss.	This reduces the availability of oxygen, increasing the production of 3-phosphoglycerate.
B	Plants synthesise large volumes of rubisco.	This may be an adaptive response to compensate for low concentrations of oxygen.
C	Processing 2-phosphoglycolate will eventually release carbon dioxide.	This will increase the rate of reduction and increase the rate of RuBP regeneration, increasing the rate of photosynthesis.
D	Rubisco is an inefficient photosynthetic enzyme.	This is because the rate of carbon dioxide reduction can be decreased by competitive binding of oxygen molecules to the active site

- 26** Which statements are acceptable parts of Darwinian evolutionary theory?

- 1 Advantageous behaviour acquired during the lifetime of an individual is likely to be inherited.
- 2 In competition for survival, the more aggressive animals are more likely to survive.
- 3 Species perfectly adapted to a stable environment will continue to evolve.
- 4 Variation between individuals of a species is essential for evolutionary change.

- A** 1, 2 and 4 only
- B** 2 and 3 only
- C** 3 and 4 only
- D** 4 only

- 27** When organochlorine insecticides such as DDT were in widespread use, mosquitoes in malarial regions developed resistance more rapidly than did houseflies in Britain.

What could account for the difference in the rates of the development of resistance?

- A** Houseflies produce more generations a year.
- B** More insecticides was used in Britain.
- C** More insecticides was used in malarial regions.
- D** Mosquitoes show fewer random mutations per generation.

- 28** Human activity often results in habitat loss. The remaining habitat in an area become fragmented forming smaller patches of habitat, through for example, construction of new roads and deforestation.

Which statements describe how a small habitat patch differs from a larger patch of the same habitat?

- 1 biodiversity decreases
- 2 competition from surrounding habitats increases
- 3 gene pool increases
- 4 populations of large animals decrease

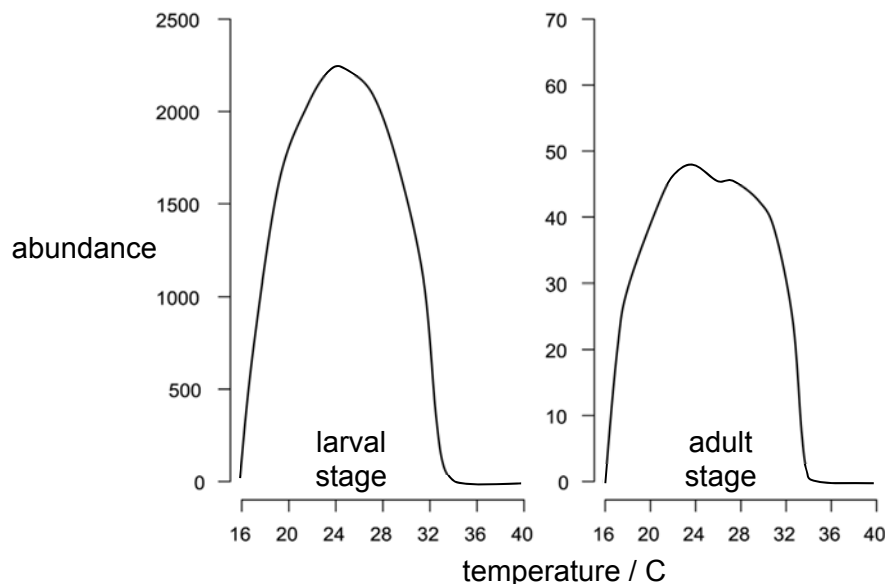
- A** 1 and 2 only
- B** 2 and 3 only
- C** 3 and 4 only
- D** 1, 2 and 4 only

- 29** Which of the following correctly shows the effects of climate change on coral reefs and associated ecosystems?

	Average number of zooxanthellae in each polyp	Mass of basal plate of hard corals	Diversity of catch from nearby fisheries
A	Decreased	Decreased	Decreased
B	Decreased	Unaffected	Increased
C	Increased	Decreased	Decreased
D	Increased	Unaffected	Increased

- 30** Malaria is caused by the protozoan parasite, *Plasmodium falciparum* (*P. falciparum*). Female *Anopheles* mosquitoes pick up *P. falciparum* in a blood meal taken from an infectious person. *P. falciparum* then go through several developmental stages before they migrate to the mosquito salivary glands. Once in the salivary glands, the parasites can be transmitted to a susceptible human host when the mosquito takes another blood meal. The time spent for the parasite to develop in the mosquito is determined by temperature.

Both *Anopheles* and *P. falciparum* are sensitive to temperature. Each stage in the life cycle of *Anopheles* mosquitoes (i.e. egg, larva, pupa and adult) is dependent on temperature, examples of which are illustrated in the following graphs.



Investigations into the effect of global warming on malaria transmission often focused on the blood meal-egg laying stage in adult females.

Which row shows the reason for and limitation use for the research?

	reason for the use	limitation of the use
A	Temperature-dependencies are not the same across the different developmental stages of the <i>Anopheles</i> mosquitoes.	Increased temperature increased larval mortality and decreased developmental speed.
B	<i>P. falciparum</i> is transmitted by adult females.	Optimum temperature for <i>P. falciparum</i> growth does not necessarily correspond to the vector's optimum temperature.
C	<i>P. falciparum</i> is transmitted by adult females.	Temperature-dependencies are not the same across the different developmental stages of the <i>Anopheles</i> mosquitoes.
D	Optimum temperature for <i>P. falciparum</i> growth does not necessarily correspond to the vector's optimum.	Increased temperature increased larval mortality and decreased developmental speed.

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